

# Single-molecule multimodal timing of in vivo mRNA synthesis

JC notes

24th March, 2026

## 1 Abstract

- Poly(A) tail measurement to distinguish transcribing vs. poly(A) transcripts.
- Splicing is much more delayed than previously thought, for at least 10 kb.
- Splicing mostly apparent only in poly(A) transcripts → they think it's delayed mostly after 3' end formation.
- Lots of m<sup>6</sup>A on unspliced RNA → deposition of m<sup>6</sup>A happens before splicing.
- m<sup>6</sup>A suppressed around exon boundaries → m<sup>6</sup>A topology should be established before EJC deposition.
- 3' end cleavage occurs multiple kbs behind transcription and is coordinated with transcription termination. The 3' end formation is recursive in many genes.
- Overall, they think that cleavage, termination and m<sup>6</sup>A deposition occur earlier than most of the splicing.

## 2 Intro

- Spliceosome assembly is known to be rapid.
- Splicing kinetics: estimates vary across methods.
- Previously, it was thought that EJC prevents m<sup>6</sup>A deposition at exon boundaries. Here, they argue that m<sup>6</sup>A is placed before splicing (and thus, before EJC placement).
  - m<sup>6</sup>A methylation machinery is known to interact with Pol II → permitting co-transcriptional methylation.

## Poly(A) notes

- Poly(A) site: canonical AAUAAA motif.
- When Pol II reaches it, it keeps transcribing.
- Protein complex cut RNA there; poly(A) polymerase (PAP) adds  $\sim 200$  As.
- Transcription termination:
  - Torpedo model: Exonuclease XRN2 start chewing the 5' end of RNA still transcribed by Pol II  $\rightarrow$  when it catches Pol II, it knocks it off DNA.
  - Allosteric model: Pol II changes conformation at poly(A) site, becomes less stable and eventually falls off DNA.

## 3 Results

### 3.1 Development of a multimodal direct pre-mRNA sequencing technique + dFORCE distinguishes endogenously polyadenylated pre-mRNA

- Need to distinguish whether pre-mRNA has a poly(A) tail:
  - pre-mRNA<sup>e</sup>: undergoing transcription.
  - pre-mRNA<sup>a</sup>: polyadenylated pre-mRNA, undergoing post-transcriptional maturation.
- They pulled Pol II with associated pre-mRNA via immunoprecipitation. The pre-mRNA<sup>a</sup> was still somehow pulled (even though it was already cleaved - otherwise it would not have the poly(A) tail); probably was somehow attached via the cleavage complex, or some chromatin components.
- They got about 90% of pre-mRNA<sup>e</sup> and 10% pre-mRNA<sup>a</sup>.
- 95% of pre-mRNA<sup>e</sup> have unspliced introns, while this is only 58% for pre-mRNA<sup>a</sup>.
- Nanopore sequencing requires poly(A) tail  $\rightarrow$  added artificially during library preparation. This *in vitro* poly(A) tail is shorter (20-80nt), so they get bimodal distribution of tail lengths, and classify by cut-off.
  - Also, the reads classified as pre-mRNA<sup>a</sup> have the start of poly(A) highly enriched by known poly(A) sites.

### 3.2 The landscape of pre-mRNA 3' polyadenylation

- They detect 42k poly(A) sites (PAS) across 11.6k protein-coding genes.
- Detected PAS mostly agreed with annotated PAS, the AAUAAA signal, and were also detected by total RNA-seq.
- About 18% of PAS were in introns (mostly downstream from canonical sites).
- Recursive polyadenylation: reads classified as pre-mRNA<sup>a</sup>, spanning between two PAS (avg. distance 531 nt). Suggesting that pre-mRNA with poly(A) tail can be cleaved again (at earlier PAS) and also polyadenylated again there.

### 3.3 Single-molecule kinetics of cleavage and termination

- Use pre-mRNA<sup>e</sup> extending beyond distal PAS. Classified as cleaved vs. uncleaved, based on their 5' end.
- They model:
  - $C_{50}$ : distance beyond PAS where 50% transcripts are cleaved.
  - $T_{50}$ : distance beyond PAS where density of Pol II active site drop to 50% of coverage downstream of PAS.
- They estimate  $C_{50}$  to 2.1kb (lower bound, because of issues with max. read length).
- Estimate of  $T_{50}$  of 2.3kb.
- When both estimates were available, the median difference was about 95 nt.

### 3.4 Discrepant evidence for co-transcriptional splicing catalysis